

MITHELESH K

Software Developer

mitheleshk3005@gmail.com — +91-6385710771 — Portfolio — LinkedIn — GitHub

Professional Summary

Software Developer with hands-on experience in MERN stack development and AI system integration. Experienced in building responsive web applications, RESTful APIs, and intelligent applications powered by Retrieval-Augmented Generation (RAG), vector search, and multi-agent architectures. Strong interest in solving real-world problems through scalable software engineering and AI-driven automation.

Technical Skills

Languages: Java, JavaScript

Frontend: React.js, Redux, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js

Databases: MongoDB, MySQL, PostgreSQL

AI/LLM: LangChain, OpenAI, Gemini, Ollama, Claude, RAG, Multi-Agent Systems

AI Concepts: Embeddings, Semantic Search, Hybrid Retrieval, Vector Databases (Chroma Cloud)

Cloud/DevOps: AWS EC2, Vercel, Docker

Tools: Git, GitHub, Postman, Cloudinary, Render

Projects

Agentic-CX — AI-Powered Customer Support Platform

Live — GitHub

Tech Stack: React, Tailwind CSS, Node.js, Express.js, MongoDB, Chroma Cloud, LangChain, OpenAI

- Built an AI-powered customer support platform to automate query handling and improve response accuracy. Architected a multi-agent AI platform utilizing a **Router Agent** to intelligently classify and route user intents between a **RAG Agent** for knowledge retrieval and an **Action Agent** for automated task execution.
- Engineered an advanced **Hybrid Retrieval** pipeline (vector + keyword search) and integrated the **Ragas evaluation framework**, achieving **86.8% Context Precision** and increasing overall answer relevancy by 30% and reducing average manual ticket resolution time by over 40%.

DevConnect — Networking Platform

Live — GitHub

Tech Stack: React, Redux Toolkit, Node.js, Express.js, MongoDB, JWT, AWS EC2, Nginx

- Built a developer networking platform using **React, Redux Toolkit, Node.js, and Express.js**, enabling profile creation, connection management, and scalable REST APIs for user interactions.
- Implemented secure **JWT authentication**, designed a scalable **connection request lifecycle** using MongoDB relationships, developed **paginated feed APIs**, and deployed the application on **AWS EC2** with **Nginx reverse proxy and HTTPS**.

HackFest — Hackathon Management System

Live — GitHub

Tech Stack: React.js, Node.js, Express.js, MongoDB, JWT, Socket.IO

- Built a full-stack platform to **digitize hackathon management**, enabling **Role-Based Access Control (RBAC)** for admins, judges, and participants with secure JWT authentication.
- Developed multi-round evaluation workflows with timeline management, team readiness tracking, complaint handling, and real-time monitoring dashboards.

Education

B.Tech in Information Technology

2023 – 2027

Bannari Amman Institute of Technology, Sathyamangalam

CGPA: 7.79/10

Higher Secondary Education

2023

Srinivasa Vidhyalaya Matriculation Higher Secondary School, Udumalpet

89.89%

Achievements

- **Theme Winner at CMR HackFest 3.0.** Secured 1st place in EdTech for building **EduGroot**(Live)
- Former Member of **National Service Scheme (NSS)** — Volunteered in awareness programs and environmental initiatives.